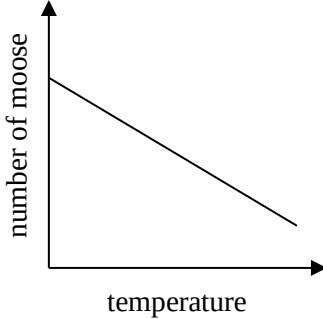


Question number	Answer	Additional guidance	Mark
9(a)(i)	extrapolation / line of best fit / calculation of mean decrease (per year)		(1)

Question number	Answer	Additional guidance	Mark
9(a)(ii)	- values read from the graph and subtracted - percentage drop calculated	$7.6 - 3.4 / 4.2$ $4.2 \times 100 \div 7.6 = 55 / 55.3 / 55.26 (\%)$	(2)

Question number	Answer	Additional guidance	Mark
9(a)(iii)	An answer that includes the following points: - temperature on the x axis (1) - number of moose on the y axis (1) {relatively / stepped} straight line sloping down from top left to bottom right (1)	ACCEPT rainfall / days of drought ACCEPT axes labelled the other way for 1 mark NB Check direction of slope if axes wrongly labelled for a CE  ALLOW a correct graph of temperature against year for 1 mark ALLOW a double y axis graph correctly labelled + line for three marks	(3)

Question number	Answer	Additional guidance	Mark
9(b)(i)	- total number of moose added up and total number of moose with 50 000 or more ticks calculated - percentage calculated to max 2 dps	214 and 41 $41 \times 100 \div 214 = 19 / 19.16 / 19.2$ CE applies if only one of the two numbers is incorrect	(2)

Question number	Answer	Additional guidance	Mark
*9(b)(ii)	Indicative content: Comment on global warming (S1) - global warming will increase the temperature of the earth's {surface / atmosphere} - winters will get warmer so less snow - winters will get shorter so snow present for fewer days Effect of change on ticks (S2) - warmer conditions decrease life cycle time - fewer ticks will die in the snow in early spring - more females to lay eggs - larvae less likely to be covered in snow in autumn - so more larvae become nymphs Effect of ticks on moose (S3) - more ticks mean larger volumes of blood removed from each moose - moose become weaker if less blood in them - moose die from lack of {nutrients / oxygen / anaemia / energy} (R) - less energy for hunting so they starve (R) - less energy for reproduction (R) - if moose lose their fur they will not be able to keep warm - moose die from the cold (R) - scratching can cause open wounds that can get infected - ticks pass on pathogens - moose die from infections (R)	Level 1 : 1 mark = description made from one section 2 marks = descriptions made from at least two sections but no links Level 2 : 3 marks = a link made between descriptions of two sections 4 marks = at least two links made between descriptions of all three sections Level 3 : 5 marks = links made between all sections with one reason (R) for moose number declining 6 marks = links made between all sections with two reasons (2R) for moose number declining (6)	

